

Lab Assignment 1

1- In this Experiment, we will identify the x86 arithmetic instructions and examine a program which will perform the following arithmetic operations:

Calculate:

$$(30_{10} + 15_{10}) * (575_{10} - 225_{10}) + 210$$

Using a calculator calculate the answer for the above arithmetic operations

Is it the same as the final answer in the AX register?

2- Run This code then show the result in memory in EMUx86:

```
DATA SEGMENT

A DW 10
B DW 20
result DW ?

DATA ENDS

CODE SEGMENT
ASSUME DS:DATA, CS:CODE

START:
MOV AX, DATA
MOV DS, AX

MOV AX, A
ADD AX, B
MOV result, AX
MOV BX, result

MOV AH, 4CH
INT 21H

CODE ENDS
END STAR
```

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